

Final Report

Anonymous Authors

Project 1

1.1 Technical Details

1.1.1 Modulation Method

Since there are three modulation method which were taught in the class, that is ASK, FSK and PSK, we first ruled out ASK, since it extremely has a high requirement on the quiet environment, what is more important, it is not recommended in the document. Intuitively we consider PSK is much more stable than FSK, therefore we chose PSK as our modulation method.

1.1.2 Programming Language

The programming language is not limited in this project, which was a big challenge for us to decide to use which one, we first tempt to use C++, as one demo uses it. But a problem occurred here that we cannot reproduce the effect of demo, the reason seems to be that the laptop of one member of our group cannot use ASIO, thus we turned to another demo, which uses Go and Jack, and this time we find its demo works well, therefore we chose Go as our Programming Language.

1.1.3 Some Implementations

At the very beginning of the transmission, we implement a warmup, which does not actually transmit any useful information, but may wake up, or to say, preheat the receiver, in the head of each header, we implemented chirp to synchronize, which contains a period of silence to avoid the impact of adjacent ones.

1.2 Challenges

1.2.1 Equipment Problems

At first we find that our code can run well with the samples directly copied from senders to receivers, but some uncontrollable problems must occur in the process of transmitting, we tried to lend other groups devices, which works well in dormitory, where is very quiet, the goodness of fit reached 1, however, when we brought the devices to the classroom, where the check is set, we find that we can never reproduce our achievement in the dormitory, which really upset us a lot.

1.2.2 Compatibility Issues

As briefly mentioned above, we noticed that ASIO could not work properly on one of our members' computers, more precisely, the demo could normally make the sound, but couldn't receive the sound it makes. The prompt words displayed in demo are as follows, 'Couldn't detect the test signal!!!Make sure there's no background noise that might be confusing it.' It is speculated that the brand of that kind of laptop, which is Lenovo, is incompatible with ASIO.

1.2.3 Simultaneous Processing of Multiple Threads

We found it difficult to solve the problem of receiving and demodulating the voice at the same time, so we played a trick, since the demand is to handle a complete part without any respond, so we designed that the receiver records everything in a buffer and then demodulates what is in it. In such way, we avoided simultaneous processing, but this also laid a great hidden danger for the future, which will be mentioned in the following text.

1.3 Suggestions

1.3.1 The Problems of Compatibility Issues

The issue that the brand Lenovo may lead ASIO to be unable to work normally should be added in the step-by-step setup instructions.

1.3.2 Availability of External Voice and Reception Equipment

Actually we didn't realize that more practical external speakers and microphones are available in the first project, as we habitually think that we can only use the laptops' own speakers and microphones. Under this kind of circumstance we couldn't meet the requirements of the project even in quiet environment such as in the dormitory, the error was very great thus we thought it might be impossible to meet the high demand, and we barely changed the code to increase its robustness. So when we arrived the classroom where is used for check, we finally found that we lost an important tool.

Project 2

2.1 Technical Details

2.1.1 Parameters for Wired Transmission

Since in the wireless transmission, the impact of noise is very large, which means that the design in the first project must take noise into consideration, which makes the sample points each part take great many, but as wires are available in the second project, such redundant number of sample points is not necessary, thus in the implementation of the second project, each signal uses only about one sixth of the sample points of the one in the first project, and each header uses only one tenth of the sample points of the one in the first project, and after some tests, it was found that the preheated part is not necessary in the transmission, and therefore it was deleted.

2.2 Challenges

2.2.1 Bottom Level Code Modification

As is mentioned above, in the first project, only one side transmission is required, that is to say, there is only one sender and one receiver, so the sender sends a complete period, so when we design the first project, we

chose to let receiver first receive the whole period, and store it in a buffer, and after the whole transmission, we let the receiver demodulate the data in the buffer. Obviously, this method is totally unavailable in the second project, as ACK is required in this project, which means that the sender is at the same time receiver, and the receiver is at the same time sender, too. In such situation, the state machine is supposed to be necessary, which is also a great challenge since we avoid such design in the first project.

2.2.2 Equipment Problems

In the process of debugging the code, great challenges occurred, that is, in many many times, the results cannot be reproduced, usually it is after the computer is in standby mode. When I met such situation at the first time, I was so confused that I thought I might mistakenly save a mistakenly modified code, as there might exist some parameters error, after such check, I repeatedly checked the entire code, again and again, but after hours and hours, I still cannot understand where is the mistake lies, at last I made every possible effort, including reconnecting the wires, trying every available wires, by the permutation, to avoid any issues with the wires, but I still couldn't reproduce what I have implemented one day ago, at this time I was almost devastated. At last, I tried to restart both of the two laptops, and this time everything works well, the fact is that I was very lucky, because the later experience told me that restart once may not solve the problem, twice or even three times may be necessary, so even after a small modification which causes some bugs, I had to repeat the previous version to ensure that it is not caused by the equipment problems. Actually even in the process of checking, I restarted both of the laptops at least three times to make what I had implemented partly work, I was too nervous after the second try, so there might be more than three times, but I cannot quite remember. Here I have to tell the students who have seen this report, you may encounter various kinds of problems in such complex projects, don't forget restarting your computer when you have no way out.

2.3 Stories

2.3.1 Actually in the process of complete this project, one member of this group had to do something else, which was also very important, so only one member was arranged to complete the second project, the truth is that this was too hard for that member, and thus we didn't do well in the second project.

2.4 Suggestions

2.4.1 Less Limits on TA's Assistance

The second project is probably too difficult, especially to those who used special method in the first project, like our group, so some help from TAs may play a significant role in this project, like theoretically explanation of the State Machine of ACK Protocol. At present, the only assistance that TA can provide is permission to view the code, which may be too limited, and make the students who played tricks on the first project taste great bitterness.

2.4.2 More and more attempts (for students)

As is mentioned before, even restarting may affect the performance of your code, so when meeting problems, don't be afraid to try whatever you can.

Project 3

3.1 Challenges

3.1.1 Unfamiliarity with essential tools

In the third project, some new tools are essential for debugging packet encapsulation and network protocols, for example, Wireshark is essential to capture and analyze the network traffic, though it was demonstrated in class, I still found it difficult to proficiently use it.

3.2 Stories

3.2.1 As the second project was not completely done, the process of the third project was progressing very slowly, and there also occurred problems with the collaboration between the team members.

3.3 Suggestions

3.3.1 Discussion Classes on essential tools

As is mentioned above, the tools such as Wireshark may be added in the discussion classes, which will be sure to be very helpful for students who are new to these tools.

Project 4

4.1 Challenges

4.1.1 The completion rate of the third project was not high, which directly affected the progress of this project.